

Directed Acyclic Graphs: Marginal Independence

STSCI / INFO / ILRST 3900: Causal Inference

22 Sep 2026

Logistics

- ▶ Aim to have Quiz 1 grades back by end of week
- ▶ Quiz 2: Oct 8

Learning goals for today

At the end of class, you will be able to

- ▶ draw a causal Directed Acyclic Graph
- ▶ enumerate edges in the graph
- ▶ read statistical dependence of nodes in the graph
- ▶ determine marginal exchangeability in the graph

After class:

- ▶ Hernán and Robins 2020 Chapter 6.1 and 6.2

Broad road map

Part I

- ▶ How do we define a causal effect? How is it different from association?
- ▶ What assumptions allow us to recover a causal effect from data? (e.g., conditional exchangeability)
- ▶ When conditional exchangeability holds, how can I recover a causal effect from data? (e.g., standardization, IPW)

Broad road map

Part I

- ▶ How do we define a causal effect? How is it different from association?
- ▶ What assumptions allow us to recover a causal effect from data? (e.g., conditional exchangeability)
- ▶ When conditional exchangeability holds, how can I recover a causal effect from data? (e.g., standardization, IPW)

Takeaway:

When conditional exchangeability holds, the ATE can be recovered by standardization or IPW

Broad road map

Part II

- ▶ How to ask good causal questions
- ▶ How can we reason about whether conditional exchangeability holds?

Broad road map

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- ▶ How can we reason about whether conditional exchangeability holds?
- ▶ Don't just condition on everything
 - ▶ Unrealistic target
 - ▶ Not a good idea, even if it possible

Broad road map

Part II

- ▶ How to ask good causal questions
- ▶ How can we reason about whether conditional exchangeability holds?
- ▶ Don't just condition on everything
 - ▶ Unrealistic target
 - ▶ Not a good idea, even if it possible

Takeaway:

Use causal graphs to clearly communicate causal assumptions and reason about conditional exchangeability

Directed Acyclic Graphs (DAGs) formalize **causal beliefs**



Causal beliefs:

- 1) Smoking may cause you to carry a lighter
- 2) Smoking may cause Heart Disease
- 3) Carrying a lighter does not cause Heart Disease

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Nodes represent random variables

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Edges represent direct causal effects

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Additional Requirement

- In this class we will think about **acyclic** graphs

Directed Acyclic Graphs (DAGs) formalize **statistical dependence**



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(Smokes, Carries Lighter) are statistically dependent
— because (Smokes) causes (Carries Lighter)

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Task. Propose a rule for when two nodes are dependent

Two nodes are dependent if and only if _____



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Path

A sequence of edges connecting two nodes without repeated nodes

Smokes $\rightarrow$ Carries Lighter Smokes $\rightarrow$ Heart Disease
Carries Lighter $\leftarrow$ Smokes $\rightarrow$ Heart Disease

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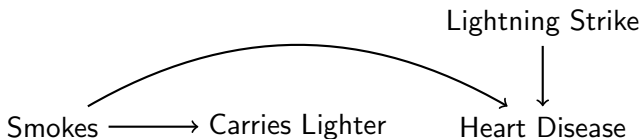
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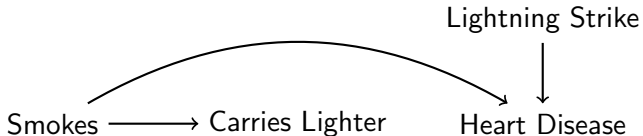
Smokes → Carries Lighter Smokes → Heart Disease
Carries Lighter ← Smokes → Heart Disease

Causal Path All edges point away from treatment
and towards the outcome

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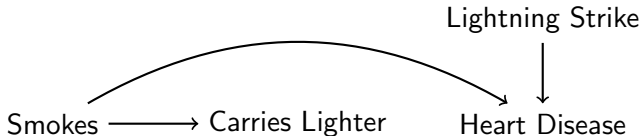
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(Lightning Strike) causes (Heart Disease)

There are no common causes of (Smokes, Lightning Strike)

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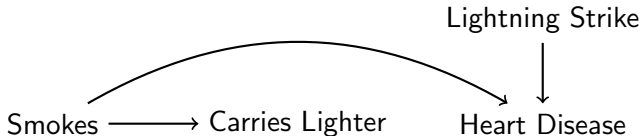


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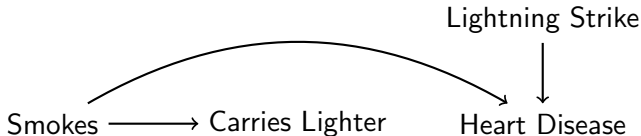
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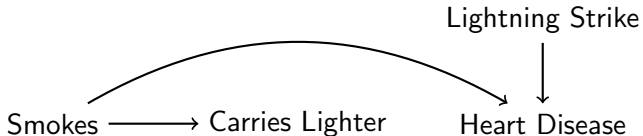
Is (Smokes) statistically related to (Lightning Strike)?

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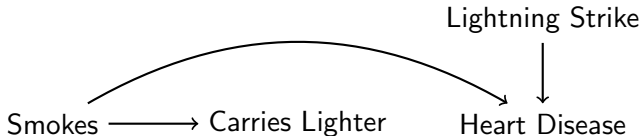
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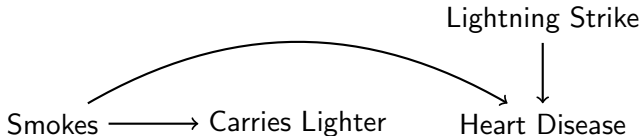
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A collider **blocks the path**.

A blocked path does not create statistical dependence.

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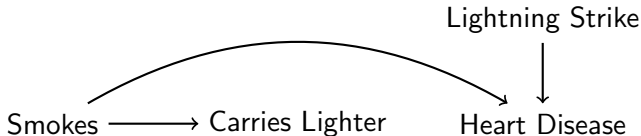
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Intuition: If two variables affect one outcome,
that does not make those two variables related

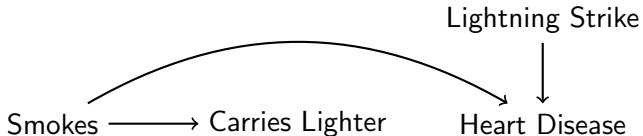
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Possible rule Two nodes are dependent if and only if
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Task. Propose a rule for when two nodes are dependent



Rule

Two nodes are dependent if and only if they are connected by an unblocked path (path with no colliders)

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Exchangeability holds if all unblocked paths between A and Y are causal paths that point from A to Y

Practical procedure

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Philosophy

1. Drawing a DAG forces you to be clear about your causal assumptions
2. Clearly think through and communicate what variables are relevant and where edges appear
3. Can argue about the DAG, but once you agree on the DAG implications for exchangeability follow

Practice

Procedure

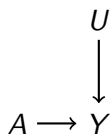
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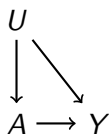
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DAG 1



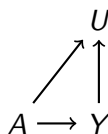
DAG 2



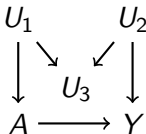
DAG 3



DAG 4



DAG 5



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